

Detour back into differential equations:

Let's explicitly solve

$$\tau \frac{dx}{dt} = -x + \underbrace{I(t)}_{\text{Time dependent}}$$

The trick is to multiply by $e^{t/\tau}$ and rearrange

$$\tau \frac{dx}{dt} e^{t/\tau} = -x e^{t/\tau} + I e^{t/\tau}$$

$$\tau \left(\frac{dx}{dt} e^{t/\tau} + \frac{1}{\tau} x e^{t/\tau} \right) = I e^{t/\tau}$$

$$\frac{d}{dt} (x e^{t/\tau}) = \frac{1}{\tau} I e^{t/\tau}$$

Integrate both sides from $-\infty$ to t .

$$x(t) e^{t/\tau} = \int_{-\infty}^t dt' \frac{1}{\tau} I(t') e^{t'/\tau}$$

$$x(t) = \int_{-\infty}^t dt' \frac{1}{\tau} e^{-(t-t')/\tau} I(t')$$

Let $K_s^{(e)}(t)$ denote the exponential kernel

$$K_s^{(e)}(t) = \begin{cases} \frac{1}{\tau} e^{-t/\tau}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Then

$$x(t) = \int_{-\infty}^{\infty} dt' K_s^{(e)}(t-t') I(t') = (K_s^{(e)} * I)(t)$$

Exponential synapses:

Recall the sub-threshold dynamics of a spiking neural network.

$$C \frac{dV_i}{dt} = -G_L (V_i - E_L) + I_i(t)$$

$$I_i(t) = \sum_{j=1}^N \underbrace{W_{ij}}_{\text{weights}} \underbrace{(K_s * \Delta_j)}_{\text{spike train}}(t) + \sum_{j=1}^M \underbrace{u_{ij}}_{\text{weights}} \underbrace{(K_s * X_j)}_{\text{spike train}}(t)$$

We said that the shape of K_s is approximately a double exponential. However, rise times can be fast, and it's often convenient to choose K_s as a single exponential.

$$K_s(t) = K_s^{(e)}(t)$$

Then we can write the synaptic current as

$$I_i(t) = \left(K_s^{(e)} * \left(\sum_{j=1}^N W_{ij} \Delta_j + \sum_{j=1}^M u_{ij} X_j \right) \right)(t)$$

and recognize it as the solution to

$$\tau_s \frac{dI_i}{dt} = -I_i + \sum_{j=1}^N W_{ij} \Delta_j + \sum_{j=1}^M u_{ij} X_j$$

where τ_s is the synaptic time constant. Our spiking neural network thus corresponds to $2N$ coupled differential equations

$$C \frac{dV_i}{dt} = -G_L (V_i - E_L) + I_i$$

$$\tau_s \frac{dI_i}{dt} = -I_i + \sum_{j=1}^N W_{ij} \Delta_j + \sum_{j=1}^M u_{ij} X_j$$

plus an integrate-and-fire mechanism that couples V_i and Δ_i in the usual way.

Firing Rate Models:

Poisson variability suggests that firing rates matter most to neural coding. Do we need to explicitly model spiking?

Let's average the current equation over Poisson variability and a small time interval Δt

$$\frac{1}{\Delta t} \int_t^{t+\Delta t} dt' \tau_s \frac{dI_i}{dt} = \frac{1}{\Delta t} \int_t^{t+\Delta t} dt' (-I_i) + \sum_{j=1}^N W_{ij} \frac{1}{\Delta t} \left\langle \int_t^{t+\Delta t} dt' \Delta_j(t') \right\rangle + \sum_{j=1}^M u_{ij} \frac{1}{\Delta t} \left\langle \int_t^{t+\Delta t} dt' x_j(t') \right\rangle$$

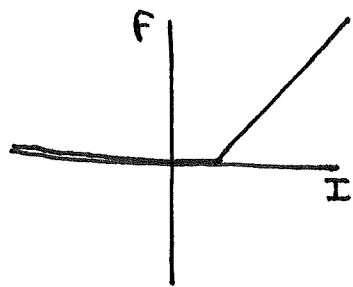
$$\frac{1}{\Delta t} \tau_s \frac{dI_i}{dt} \Delta t = -\frac{1}{\Delta t} I_i \Delta t + \sum_{j=1}^N W_{ij} \frac{1}{\Delta t} \kappa_j \Delta t + \sum_{j=1}^M u_{ij} \frac{1}{\Delta t} x_j \Delta t$$

where x_j is the firing rate of the j^{th} input neuron, and we assumed that Δt is small enough that I_i and dI_i/dt do not change appreciably in t to $t+\Delta t$.

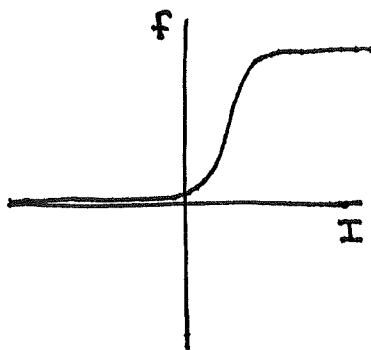
This becomes a complete firing rate model if we compute κ_i from I_i with an f - I curve.

$$\tau_s \frac{dI_i}{dt} = -I_i + \sum_{j=1}^N W_{ij} \kappa_j + \sum_{j=1}^M u_{ij} x_j \quad (I)$$
$$\kappa_i = f(I_i)$$

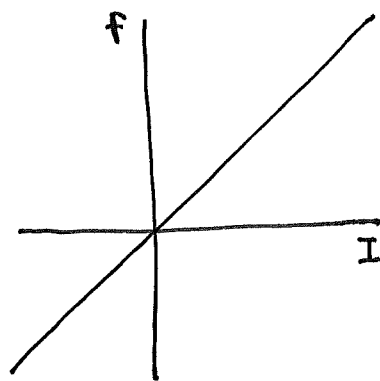
Typical choices for f - I curves



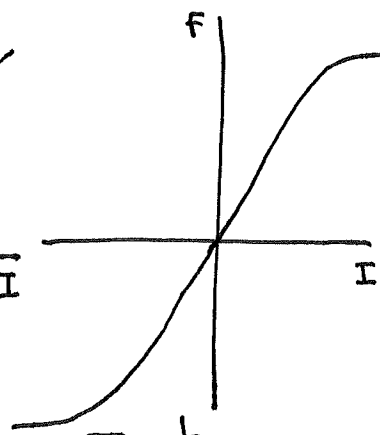
Threshold-linear



Sigmoidal



Linear



Tanh

A more realistic model allows firing rate changes to take some time.

$$\left. \begin{aligned} \tau_s \frac{dI_i}{dt} &= -I_i + \sum_{j=1}^N W_{ij} h_j + \sum_{j=1}^M u_{ij} x_j \\ \tau_h \frac{dh_i}{dt} &= -h_i + f(I_i) \end{aligned} \right] \quad (\text{II})$$

where τ_h is the exponential time scale of the firing rate response.

If $\tau_s \ll \tau_h$, we can replace I_i with its steady-state

$$\tau_h \frac{dh_i}{dt} = -h_i + f\left(\sum_{j=1}^N W_{ij} h_j + \sum_{j=1}^M u_{ij} x_j\right) \quad (\text{III})$$

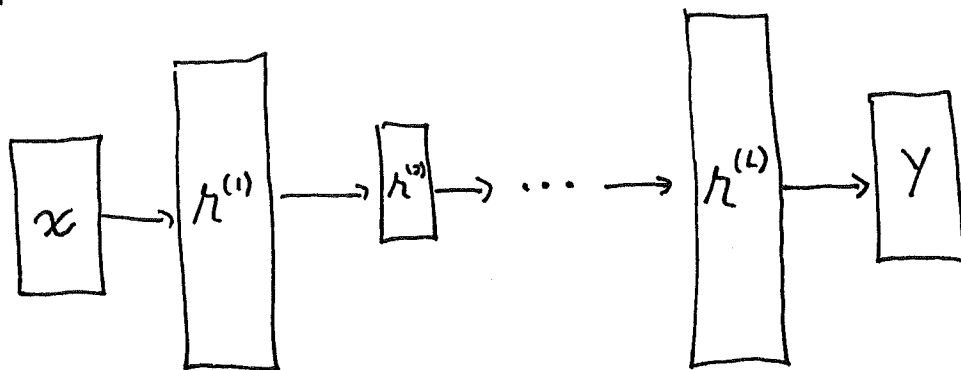
We thus have three versions of the firing rate equations. (II) is most general but rarely used in practice. (I) assumes fast firing rate changes. (III) assumes fast synapses and is most commonly used.

Feedforward networks:

If $W_{ij} = 0$, then the steady-state firing rates are simply

$$h_i = f\left(\sum_{j=1}^N u_{ij} x_j\right) \quad \text{or} \quad \underbrace{h = f(u x)}_{\text{vector form}}$$

This can be repeated many times to form layered architectures



x : input layer
 $h^{(l)}$: hidden layers
 y : output layer
 L : #hidden

This layering is the foundation of deep neural networks.